

Yu-Hsuan(Michael) Huang

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EDUCATION

University of Michigan – Ann Arbor
M.S. in Robotics

Ann Arbor, USA
Sep 2026 – Dec 2027 (expected)

National Taiwan University
M.S. in Mechanical Engineering

Taipei, Taiwan
Sep 2015 – Jul 2017

National Taiwan University
B.S. in Civil Engineering

Taipei, Taiwan
Sep 2012 – Jul 2015

CORE COMPETENCIES

- Robot Learning (Imitation Learning)
- Motion Planning & Control
- System Integration

PROFESSIONAL EXPERIENCE

Shin Fang Co., Ltd.
Robotics Engineer

Taipei, Taiwan
Oct 2023 – Aug 2026

- Established an end-to-end imitation learning pipeline on a PyTorch-based framework, covering data collection, training, and deployment in simulation and on hardware, achieving autonomous behavior in real-world tests.
- Engineered a low-latency teleoperation system, working on both hardware and software architecture, enhancing control algorithms, and testing multiple strategies, which enabled smooth and accurate data collection.
- Built a modular planning and control framework for a dual-manipulator kitchen robot, decoupling planners, end-effectors, and execution target so each could be swapped freely, on real hardware or its Isaac Sim digital twin.
- To scale the robot to new dishes, decomposed complex cooking processes into reusable task-planning subtasks, so new workflows were assembled from existing skills instead of rewritten.
- Created a vision-based bin-picking function with a foundation-model backend, improving picking consistency in clutter.
- Integrated a custom electronic stove control unit for automated flame control, improving system responsiveness and task stability under real-world kitchen conditions; named inventor on US patent application 2026/0110436 A1.
- Enabled commercialization of the dual-arm robot on a low-cost hardware platform (~1/5 the cost): Assembled and validated it on-site with the hardware partner, then ported the learning pipeline to it.
- Helped establish CI/CD practices in the company's early stage, with daily source and container builds and automated checks gating every merge, making multi-developer collaboration scalable.

Synopsys Taiwan Co., Ltd.
R&D Engineer

Taipei, Taiwan
Jul 2022 – Jul 2023

- To keep IC Validator, Synopsys's physical verification tool, release-ready, triaged daily regression failures and pinpointed the version behind each, helping the team meet industry-standard benchmarks and earn the Software Quality Award.

Brogent Technology, Inc.
Advisory Software Engineer

Kaohsiung, Taiwan
Jan 2022 – Jun 2022

- Independently designed and built the full hardware and software stack for an autonomous mobile robot, delivering a working prototype that demonstrated technical feasibility to executives.

International Center of Excellence in Intelligent Robotics and Automation Research
Research Specialist

Taipei, Taiwan
Dec 2018 – Oct 2021

- Designed gravity compensation algorithms to enable intuitive teach-and-play control on an industrial robot arm, improving ease of use and increasing operator safety in manual programming scenarios.
- Formulated a deterministic real-time trajectory generation algorithm running at a 1 kHz control loop on IntervalZero RTX64, ensuring smooth, responsive robot movement.
- Integrated a robot arm with an AOI machine, deploying a pick-and-place cell with optical sensors to automate spray-gun quality checks and reduce human inspection errors.
- Built a 3D-printed six-axis robot arm from scratch with 1 kg payload, assembling mechanical components, followed by developing a motion planning framework to enable essential robotic arm movements and control.

TECHNICAL SKILLS

Languages: C++, Python

Robotics Frameworks: ROS, ROS 2, MoveIt 2, ROS 2 Control, BehaviorTree.CPP, LeRobot

Development & Real-Time: Linux, Docker, Git, CI/CD, IntervalZero RTX64

Simulation & Design: NVIDIA Isaac Sim, SolidWorks, MATLAB

PUBLICATIONS

- **Y.-H. Huang**, H.-C. Hsu, Y.-F. Chen, T. Liu, and C.-P. Yang, "Design and modeling of a novel 6-speed dual clutch transmission system," *IEEE Int. Conf. on Mechatronics and Automation (ICMA)*, 2017, pp. 141–146. [DOI]